

Piezoelectric Properties of Electrospun Polymer Nanofibers and Related Energy Harvesting Applications

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Electrospinning (ES) methods that can produce piezoelectricity in polymer nanofibers have attracted tremendous research attention. These electrospun polymer nanofibers can be employed for sensors, energy harvesting, tissue engineering, and filtration applications. This paper reviews the performance of a variety of electrospun piezoelectric polymer nanofibers produced by different ES methods, including near-field electrospinning and conventional far-field electrospinning methods. Herein, it is described how the ES method can affect the piezoelectric properties of various polymer nanofibers, including poly(vinylidene difluorine), poly(vinylidene fluoride-trifluoroethylene), nylon 11, poly(L-lactic acid), and poly(α -benzyl-L-glutamate). Due to the varied matrix structures of piezoelectric polymer nanofibers, the ES method may conduct variable effects on the piezoelectric properties of polymer nanofibers. After characterizations by X-ray diffraction, Fourier transform infrared spectrum, dielectric spectra, and piezoelectric coefficient measurements, it is found that the piezoelectric properties of the polymer nanofibers can be significantly affected by the ES parameters. Most of previous review articles focus on the output performance of electrospun polymer nanofibers. A detailed description of how different ES methods affect the piezoelectricity of polymer nanofibers is still lacking. In this review paper, the basic principle behind ES methods and the way in which different ES methods affect the properties of polymer nanofibers are examined.

1. Introduction

Over the last decade, electrospinning (ES) has attracted significant research and engineering attention due to its applications in tissue engineering, biomedical devices, and energy harvesting devices.^[1–3] During the ES process, a polymer solution is driven by a syringe pump through a syringe needle in an ES setup. The polymer solution forms a Taylor cone on the tip of ES nozzles, which can be stretched by an electrostatic force to form polymer fibers on a static or rotational collector. During this process, the dipoles or functional group inside nanofibers can be rearranged by the electrostatic force or the shear force during the ES. For piezoelectric polymer nanofibers, including poly(vinylidene difluorine) (PVDF) and poly(L-lactic acid) (PLLA), piezoelectricity on the electrospun polymer nanofibers can be produced during this process. Since the 2010s, a large number of review papers have been published in this field to reveal the properties of electrospun polymer nanofibers. However, most of these papers focus on discussing the output performance of electrospun polymer nanofibers

and related applications. The fundamental principle of this technology has yet to be understood, including how different ES methods affect the piezoelectric properties of various piezoelectric polymers with various matrix structures. In this review paper, we will review the ES process using near-field electrospinning (NFES) and conventional far-field electrospinning (FFES) methods to compare the piezoelectric properties of polymer nanofibers, including PVDF, nylon (polyamide) 11, PBLG (poly(γ -benzyl α , L-glutamate)), and PLLA (poly(L-lactic acid)) to fully understand the fundamental science behind these technologies.

Since the 1960s, piezoelectric polymers have been of great interest in research and engineering because they can be employed in a broad range of applications, including transducers, loudspeakers, sensors, and energy harvesting devices.^[4–10] However, their complicated fabrication process limits their application in many respects. Therefore, a simple, scalable, cost-effective material fabrication technique is required to produce high-quality and low-cost piezoelectric polymers. In this report, we will review the fundamental science behind this technology by comparing the piezoelectric properties of various polymer nanofibers produced using ES methods and also review them for energy harvesting

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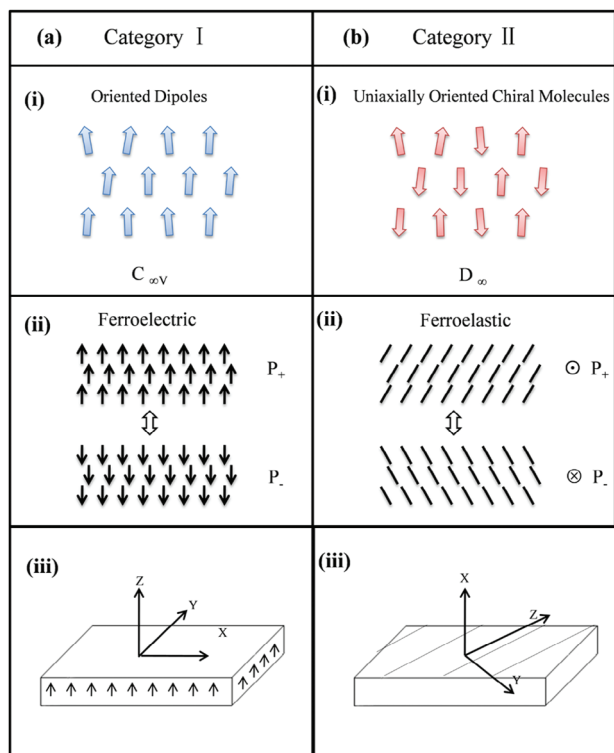


Figure 1. Schematic drawings of i) dipole orientation, ii) molecular orientation, and iii) Cartesian coordinate systems definition for category I and category II.

and sensor applications. Depending on the interaction between the electric field and dipole orientation in the polymer, the ES method may show various effects on the piezoelectric properties of the electrospun polymer nanofibers.

We believe that a full understanding of how different ES methods affect the piezoelectric properties of polymer nanofibers can benefit the mass production of electrospun piezopolymer nanofibers and related research in this field. It will help researchers understand and select an appropriate method for producing piezopolymer nanofibers with high piezoelectric properties.

1.1. Piezoelectric Polymer Categories

Piezoelectric polymers can be classified into two categories.^[11,12] A representative material for category I is PVDF, as the schematic shown in Figure 1a, which possesses symmetrical structures of $C_{\infty v}$, in which the dipoles in the material are perpendicular to the main carbon chain. As illustrated in Figure 1a, the Class I piezoelectric polymer matrix has a piezoelectricity in the thickness (d_{33}) and length (d_{31}) of the material. The materials in category II are mostly chiral polymers, including PBLG, PLLA, and deoxyribonucleic acid (DNA), which normally show a symmetrical structure of D_{∞} .^[11] Because the dipoles are aligned antiparallel in Figure 1b, the biopolymer materials typically exhibit piezoelectricity along the shear direction (d_{14}) and there is no piezoelectricity along the thickness direction of the material.

As shown in Figure 2a, PVDF material mainly possesses three main phase conformations, including the α phase, β phase, and γ phase.^[13] For a PVDF film fabricated using a solution casting method, it normally shows a nonpolar α phase. After the PVDF film is uniaxially stretched, it becomes a polar β phase. PVDF-TrFE material is synthesized using vinylidenedifluoride (VDF) with trifluoroethylene (TrFE) monomers to produce piezoelectric polymers PVDF-TrFE with direct ferroelectric phases (β phase). Due to the advantage of the direct polar phase (β phase) without a stretching process, PVDF-TrFE polymers are employed more widely in a broad range of applications, including sensors, biomedical devices, and transducers.^[14–16] In addition, a PVDF-based terpolymer PVDF-TrFE-CFE (or PVDF-TrFE-CTFE) was synthesized using three monomers, including VDF, TrFE, and chlorofluoroethylene (CFE) or chlorotrifluoroethylene (CTFE) to create a relaxor ferroelectric polymer PVDF-TrFE-CFE or PVDF-TrFE-CTFE, which generally shows a weak polar phase (γ phase).^[17] The terpolymer shows a relatively large dielectric constant of ≈ 60 and a large strain (3% at 100 MV m⁻¹), which shows great potential in applications of actuators, and high energy density capacitor applications.^[17]

Recently, with the growth of miniature and nanoscale devices, electrospun PVDF nanofiber-based devices have attracted tremendous research favor due to their potential applications in energy harvesting, self-powered devices, and wearable devices.^[18–20] Due to the interaction between the electrostatic force and electric field during the ES process, the PVDF nanofibers can be stretched, and the dipoles inside the PVDF nanofibers can be realigned along the electric field direction, which can introduce piezoelectricity in PVDF nanofibers. We will discuss this material in detail in the upcoming paragraph.

Since the 1980s, polyamide polymers (commonly named: nylon) have been studied due to their relatively large piezoelectric coefficient at high temperature.^[21–23] Since then, they have been widely used in applications of sensors, actuators, and air filters.^[24–27] Polyamide possesses a relatively large piezoelectric coefficient along the direction of the film thickness. As the schematic drawing shown in Figure 2b, the N–H and C=O groups between the monomers of $[\text{NH}(\text{CH}_2)_n\text{CO}]_n$ in the adjacent main chains of nylon can form strong hydrogen dipoles. An even-numbered nylon, such as nylon 6, contains 5 repeating units of NH_2 , which induces the N–H...O=C hydrogen bonds between the adjacent main chains, arranging them in an antiparallel direction. Thus, the polarity of the hydrogen bonds will cancel each other out, and nylon 6 shows no piezoelectric property. Odd-numbered nylon has an even number of (CH_2) units between the N–H and C=O groups, and the structure of the polymer chain prefers to arrange the hydrogen bonds in a parallel direction. Thus, odd-numbered nylon, such as nylon 11, possesses a piezoelectric coefficient of 2.5 pC N⁻¹ at room temperature. Nylon 11 possesses various crystallinity conformations, such as α , α' , δ , δ' , γ and γ' .^[27] Newman et al. found that γ -phase nylon 11 is easily polarized to show a large piezoelectric coefficient compared to that of the α -phase.^[23] Schenbeim et al. found that the piezoelectric coefficient d_{31} of nylon 11 reached 12 pC N⁻¹ at 90 °C.^[28]

Poly(γ -benzyl α , L-glutamate) (PBLG) is one of the representative materials belonging to Group II.^[11] As shown in Figure 2c, PBLG is a liquid crystal polymer with an α helical structure, where hydrogen bonds along the direction of the helical chain

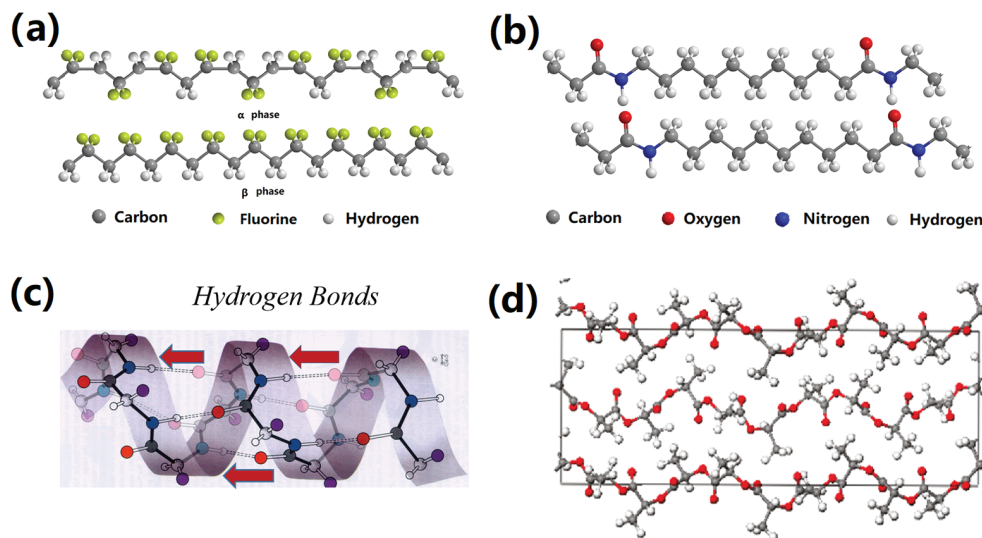


Figure 2. Schematic drawings of the molecular structures of a) PVDF, b) nylon11, c) PBLG, and d) PLLA.

are arranged in antiparallel directions.^[29] However, due to the helical structure of the PBLG film, the polarity along the dipole direction in PBLG cancels each other out, and this material only shows piezoelectric properties along the shear direction (d_{14}). In this review paper, we will show how the ES process can affect the dipole direction of PBLG nanofibers to show a piezoelectric property along the d_{33} direction.

As a widely used biopolymer, PLLA is a piezoelectric polymer that is synthesized from L-lactide monomers, which can be extracted from a variety of renewable sources, including corn and rice. Due to the advantages of biodegradability and the relatively large piezoelectric coefficient in PLLA material, it has been widely used in applications of tissue engineering, biomedical devices, and energy harvesting devices.^[30–32] As shown in Figure 2d, if the PLLA material is stretched at a low temperature, it possesses a 10_3 helical structure (α phase). Alternatively, under high-temperature stretching, it can possess a 3_1 helical structure (β phase).^[33–36] However, limited attention was paid to studying the piezoelectric properties of PLLA nanofibers. In this review article, we will discuss how the ES process can produce piezoelectricity along the dipole direction (d_{33}) in electrospun PLLA nanofibers.

1.2. ES Methods

In general, ES methods can be divided into an NFES method and a conventional FFES method.^[19,37] For the NFES method, the distance between the needle and the collector in the ES setup is normally less than 1 mm. Therefore, the strong electric field acting on the polymer solution can polarize the produced nanofibers during the solvent evaporation process.

The FFES method is one of the most widely used ES methods to produce large quantities of polymer nanofibers. During the FFES method, the gap distance between the syringe needles and fiber collectors can be from 5–25 cm. Depending on the way of collecting fibers, the FFES methods can be divided

into two subgroups. Subgroup I is the FFES method with a rotating collector (FFES-R) to align nanofibers along the ES direction. Subgroup II is the FFES method with a static collector (FFES-S) to collect nanofibers in a random form. The difference between these two methods is the interaction between the electric field and dipole orientation during the ES process. In the FFES-R method, the electric field is parallel to the ES direction. In the FFES-S method, the electric field is perpendicular to a static target. These two methods may have different effects on the piezoelectric properties of the nanofibers.

2. Energy Harvesting Using Piezoelectric Polymer Nanofibers

2.1. PVDF Nanofibers from the NFES Method

In 2010, Lin's group in Berkeley for the first time produced PVDF nanofibers using the NFES method for nanogenerator applications.^[38] As shown in Figure 3, the PVDF nanofibers were deposited on a cyclic olefin substrate using the NFES method, and two 3 m Al conductive tapes were attached on both sides of a single fiber of PVDF to act as the electrodes with a gap distance of ≈ 100 –600 μm . During the bending test, the PVDF nanofibers showed an output signal with a V_{oc} (open-circuit voltage) of ≈ 6 V and an I_{sc} (short circuit current) of 0.5–3 nA when applying a stress or a bending force to the fiber. Finally, the calculated energy conversion efficiency was 12.5% for the PVDF nanogenerator, which was much higher than that for PVDF films. During this ES process, a DC electric field (≈ 1 kV) was applied between the PVDF nanofibers and the electrode. Since the distance between the needle and the target in the NFES method is very short (< 1 mm), a strong electric field applied on them may synergistically align the dipoles along the thickness direction of the PVDF nanofibers. This may be the reason why the relatively strong piezoelectric property in the electrospun PVDF nanofibers can be produced using the NFES method.

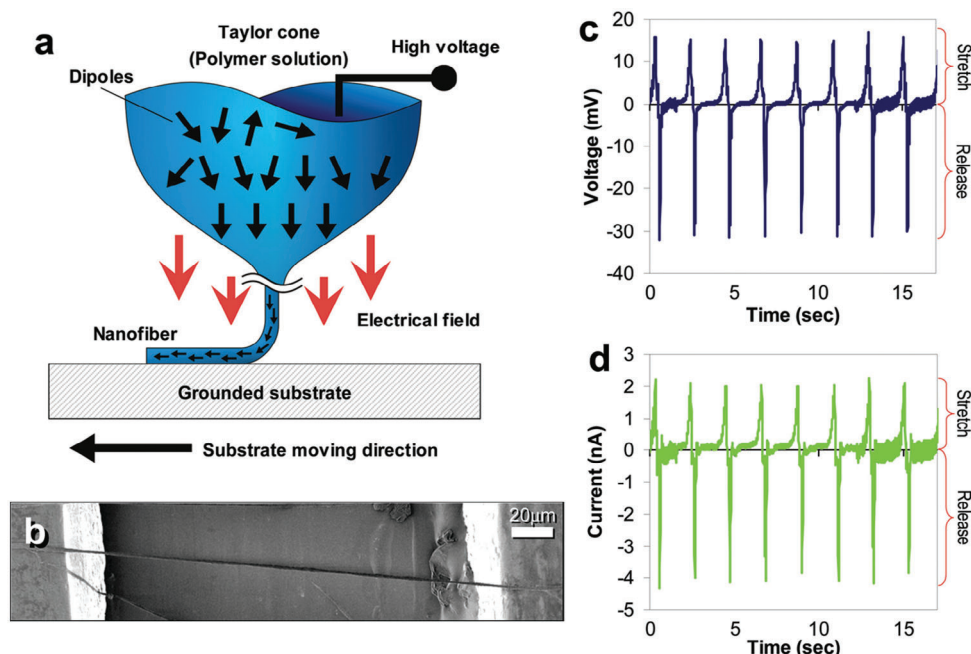


Figure 3. Piezoelectric PVDF nanogenerator. a) Near-field electrospinning (NFES) combining direct-write, mechanical stretching, and in situ electrical poling to create and place piezoelectric nanogenerators onto a substrate. b) Scanning electron microscopy (SEM) image of a nanogenerator composed of a single PVDF nanofiber, two contact electrodes, and a plastic substrate. c) Output voltage measured with respect to time under an applied strain at 2 Hz. d) Output current measured with respect to time under an applied strain at 2 Hz.^[38] Reproduced with permission.^[38] Copyright 2010, American Chemical Society.

2.2. Aligned PVDF Nanofibers from the FFES Method

In 2008, J. Andrew and D. Clarke found that ES is a simple technology to produce a polar β phase in PVDF nanofibers directly from a polymer solution.^[39] They compared the Fourier transform infrared (FTIR) spectra for electrospun PVDF fibers with different diameters and found that the intensity of the absorption peak at 839 cm^{-1} (β phase) was much higher than that of the PVDF film directly from a solution casting method. Moreover, in 2011, Hansen et al. fabricated PVDF nanofibers using the FFES method, and used a 20 kV DC voltage between the two electrodes (1 mm in thickness) to pole the PVDF nanofibers after the ES.^[40] The authors claimed that the extra poling step after the ES process is necessary to produce piezoelectric properties in the PVDF nanofibers. After the characterization, the PVDF nanogenerator generated an output voltage of $\approx 30\text{ mV}$ and 0.3 nA current. Finally, the PVDF nanogenerator was combined with a biofuel cell to drive ZnO optical devices and the data are shown in Figure 4. This is the first time that a PVDF nanogenerator has been integrated with a biodevice-based nanogenerator for a device application. We believe that the ES process can elongate PVDF nanofibers a few times (<4 times) to benefit the formation of the β phase conformation. In this case, the electric field is parallel to the main carbon chain of the PVDF nanofibers. Thus, it is very challenging to align the dipoles perpendicular to the main carbon chain. Therefore, a poling step was necessary to produce a relatively large piezoelectric coefficient in electrospun PVDF nanofibers. In 2011, Farrar et al. investigated electrospun PVDF nanofibers using the FFES method with a rotating collector.^[41] They found that the electrospun PVDF

nanofibers produced in this manner exhibited a very weak piezoelectric property. This result is consistent with the result from Hansen. In 2021, Zhang et al. compared the piezoelectric properties of electrospun PVDF nanofibers produced using various ES methods. They found that PVDF nanofibers produced using the NFES method showed the highest output current (75.63 pA cm^{-2}), higher than that of PVDF nanofibers produced using the FFES method.

2.3. Random-Aligned PVDF Nanofibers Using the FFES Method

To produce piezoelectric polymers in a random pattern, the needle in the FFES method is normally perpendicular to a static collector, and the collector or the needle is moved along the X or Y direction. In 2011, Fang et al. investigated randomly oriented PVDF nanofibers fabricated by the FFES method. The PVDF nanofibers were sandwiched between two Al electrodes for the device measurement.^[43] After an X-ray diffraction (XRD) characterization, the device exhibited a large percentage of β -phase conformation than that of a film fabricated using a solution cast method. This is mainly attributed to the long travel distance during the FFES process, in which the shear force and the electric field synergistically applied to the nanofibers to stretch and pole the fibers along the direction of the electric field. The PVDF device showed a V_{oc} of $\approx 2\text{ V}$ and an I_{sc} of $\approx 2\text{ }\mu\text{A}$. The authors declared that the mechanism for this PVDF membrane nanogenerator was not yet clear. However, we believe that the electric field direction in this case is always perpendicular to the PVDF nanofiber surface, which can facilitate the alignment of the PVDF

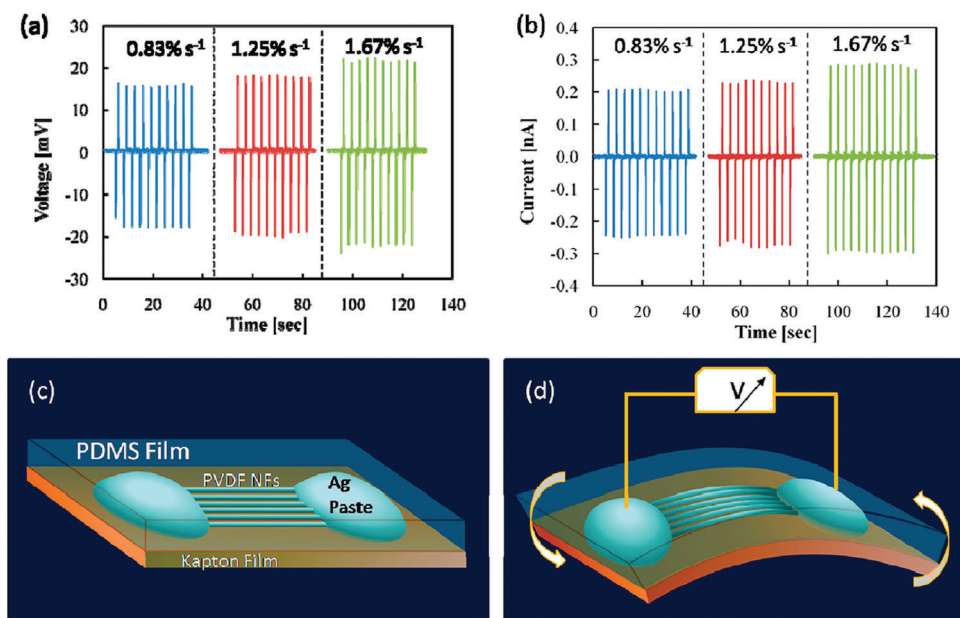


Figure 4. a) Open-loop voltage and b) short-circuit current of a PVDF NG as a function of strain rate with a maximum strain of $\approx 0.05\%$. c) PVDF NFs on a Kapton substrate, with both ends bonded with silver paste and the entire device encapsulated with polydimethylsiloxane. d) Mechanical bending of the substrate creates tensile strain and a corresponding piezoelectric field along the fibers that drive electrons through an external load back and forth following the cycled mechanical action.^[40] Reproduced with permission.^[40] Copyright 2010, American Chemical Society.

nanofibers along the thickness direction. In addition, the orientation process over the long travel distances in the FFES method can also benefit the stretching of PVDF nanofibers to show piezoelectricity along the thickness direction. In 2011, Wang et al. reported a force sensor consisting of an ITO-coated plastic film as the top flexible electrode, an ITO-coated glass as a rigid bottom electrode, and a randomly distributed PVDF fiber bundle as the middle layer. The response signal of the device depended on the frequency of the deformation, and a high sensitivity of 42 mV N^{-1} was achieved.^[44] However, we believe that the solvent remaining on the PVDF nanofibers in the static collector may dissolve the polymer nanofiber and cause a random reorientation of the dipoles in the materials, which can damage the piezoelectric properties of electrospun PVDF nanofibers produced using a static collector.

2.4. Electrospun PVDF-TrFE Nanofibers

As explained in the previous paragraph, owing to the additional TrFE monomers in PVDF-TrFE materials, the PVDF-TrFE film can directly possess a β -phase conformation after a solution cast method. In addition, for electrospun piezopolymer nanofibers, PVDF-TrFE nanofibers can generate a direct ferroelectric polar phase (β phase) during the ES process without an external stretching process.

In 2011, Mandrel et al. fabricated PVDF-TrFE nanofibers using the FFES method with a cylindrical drum as a collector. The measured FTIR spectra of the as-spun PVDF-TrFE nanofibers indicated that the polar group CF_2 was perpendicular to the main carbon chain. Compared with the electrospun PVDF-TrFE nanofibers after a heat treatment, the as-spun nanofibers

showed a peak-peak voltage of $-0.2 + 0.4 \text{ V}$, and the heat-treated nanofibers showed values of almost zero. However, the I_{sc} (short circuit current) of the device was not shown in the paper, we believe that it may be very small.^[45]

In 2015, Beringer et al. investigated electrospun PVDF-TrFE nanofibers for biological sensor applications.^[15] In their investigation, PVDF-TrFE nanofibers were fabricated using the FFES method with a static copper collector. To test the PVDF-TrFE nanogenerator, a PZT-based (lead zirconate titanate) cantilever was used to apply a force of 8 mN on the PVDF-TrFE nanofibers. As the data presented in Figure 5, the measured V_{oc} of the nanogenerator was $\approx \pm 0.4 \text{ V}$. After biocompatibility tests of the nanogenerator, it was found that all HeLa cells adhered to the nanogenerator with extended morphology.

Similarly, in 2015, Lei et al. also proved that a relatively low piezoelectric property was produced in the aligned PVDF-TrFE nanofibers produced by the FFES method with a rotating collector.^[47] They compared the randomly aligned PVDF-TrFE nanofibers with the aligned PVDF-TrFE nanofibers and the experimental results demonstrated that the randomly aligned PVDF-TrFE nanofibers could generate a V_{oc} of $\approx 200 \text{ mV}$, which was much larger than that from the aligned PVDF-TrFE nanofibers produced using the FFES-S (the FFES with a static collector) method. This may be attributed to the ES process of the random-aligned PVDF-TrFE nanofibers, in which the electric field is always perpendicular to the polymer nanofibers during the ES process.

In 2016, Ico et al. fabricated a cantilever based on electrospun P(VDF-TrFE) nanofibers (active layer) with a polyimide film (PI) as a passive layer. They accurately controlled the concentrations of the PVDF-TrFE solution to change the diameter of nanofibers from less than 100 to 1000 nm . The piezoelectric force

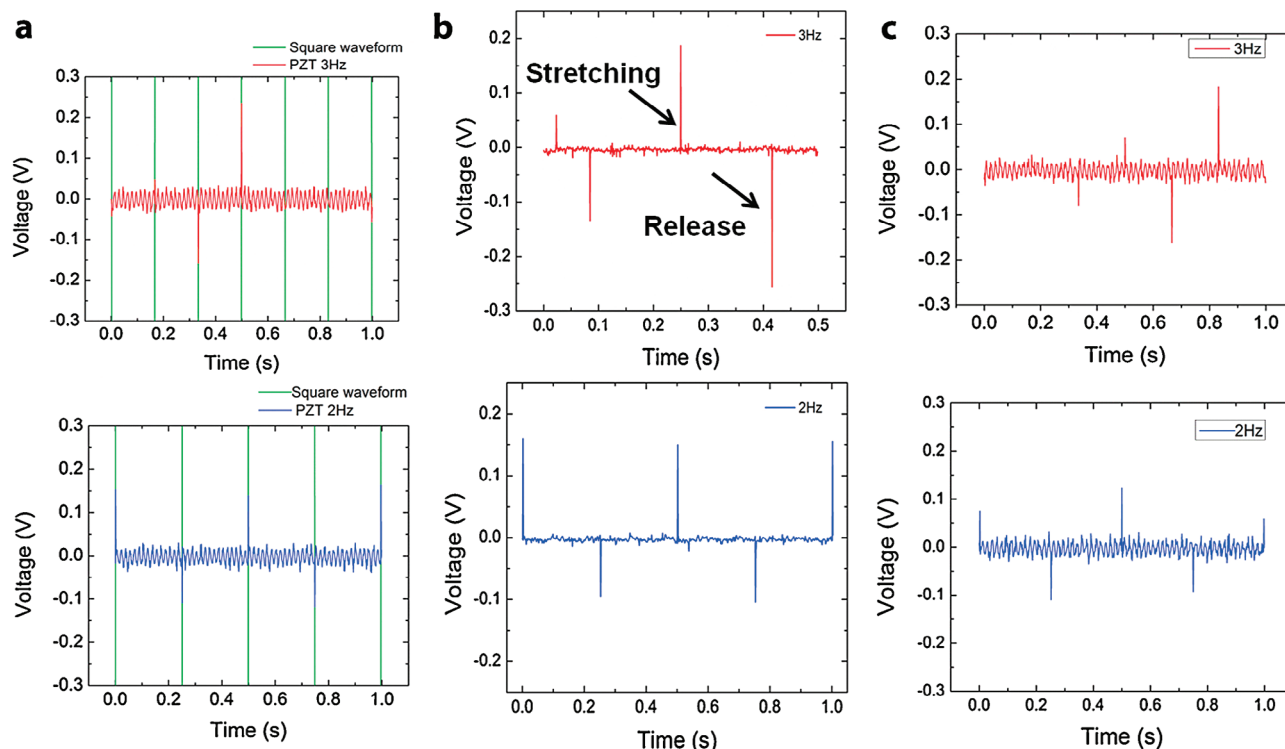


Figure 5. Representative piezoelectric response of nanogenerators in both Methyl Ethyl Ketone (MEK) solvents. a) Square waveform overlaid with voltage versus time output of PVDF-TrFe in MEK at both 2 and 3 Hz frequencies. b) Characteristic nanogenerator response of PVDF-TrFe fibers in MEK solvent. c) Characteristic nanogenerator response of PVDF-TrFe fibers in MEK solvent.^[15] Reproduced with permission.^[15] Copyright 2014, Elsevier B.V.

microscope measurement results demonstrated that the piezoelectric constant (strain constant) of the nanofibers was significantly increased as the fiber diameters were reduced to less than 500 nm.^[48]

In 2018, Wang et al. investigated electrospun PVDF-TrFE nanofibers for cell proliferation applications.^[46] They found that the PVDF-TrFE nanofibers fabricated by the FFES method exhibited an improved β -phase conformation. However, during this process, because the direction of the electric field was parallel to the ES direction, the dipoles inside the PVDF-TrFE nanofibers were minimally aligned along the thickness direction of polymer nanofibers. Therefore, an extra poling step was necessary to improve the piezoelectric property in the PVDF-TrFE nanofibers. After the poling process, the electrospun PVDF-TrFE nanofibers exhibited a piezoelectric coefficient d_{31} of 15 pC N⁻¹. The piezoelectric property of the PVDF-TrFE electrospun nanofiber after the poling process was similar to that of stretched PVDF-TrFE films. Finally, the PVDF-TrFE nanofibers were measured by a lab-designed setup that could increase a proliferation rate of fibroblast cells by $\approx 60\%$.

In 2020, Calavalle et al. studied the piezoelectric properties of electrospun PVDF-TrFE nanofibers using piezoelectric force microscope and Kelvin probe force microscopy. They clarified that the electric current of electrospun fibers was attributed to the accumulated negative space charge in the fibers, and the negative triboelectric surface charges were generated from the contact between the probes and the metal electrodes of the nanofibers.^[49]

However, in 2021, Zhang et al. compared the piezoelectric coefficient d_{31} of the randomly aligned PVDF NFs and the aligned PVDF NFs produced using the FFES method. They found that the piezoelectric coefficient d_{31} of the aligned PVDF NFs is 65.8% higher than that of the randomly aligned PVDF NFs. This means that the piezoelectric properties of PVDF NFs fabricated using the FFES method heavily depend on the fabrication process.^[42]

2.5. Electrospun Nylon Nanogenerator

As explained in the previous paragraphs, odd-number nylon shows a relatively large piezoelectric coefficient at high temperature. In 1959, Slichter indicated that all trans conformation (γ phase) in the nylon 11 film induced a net dipole moment perpendicular to the direction of the main carbon chain, providing piezoelectric properties along the d_{33} and d_{31} direction in the nylon film.^[50]

Unlike the short hydrocarbon chain, polyamides with longer hydrocarbon chains like nylon-11 cannot be dissolved in formic acid (FA) at room temperatures. Moreover, these polymers are actually resistant to most commonly used solvents, including strong mineral acids, organic acids, and strong oxidizing agents. In 1990, Shi and White^[51] produced nylon-11 fibers using a melt spinning method, which was spun from a capillary rheometer with a 0.84 mm-diameter die (L/D of 19) at 220, 240, and 250 °C. In 2007, Behler et al.^[52] produced Nylon 11 nanofibers with a

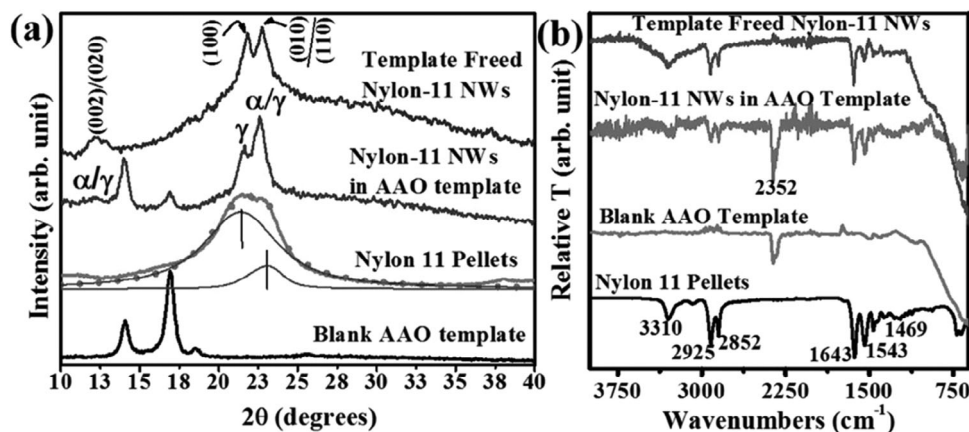


Figure 6. a) XRD patterns of nylon-11 nanowires synthesized by template wetting before and after template removal along with a comparison to the parent nylon-11 pellets. The XRD pattern of a blank AAO template is shown for reference. b) Room temperature FTIR spectra of nylon-11 nanowires showing prominent FTIR modes of the α and γ phases of nylon-11. FTIR spectra of a blank AAO template and commercial nylon-11 pellets are also shown for reference.^[54] Reproduced with permission.^[54] Copyright 2016, Wiley-VCH.

mixture solvent of FA/DCM (formic acid/dichloromethane) at different solution concentrations using the FFES method. In 2008, Dhanalakshmi et al. successfully produced γ -phase nylon 11 nanofibers using an ES process under two different concentrations of FA solutions.^[8] They claimed that the formation of γ -phase of nylon fibers was attributed to the high elongation rate during the ES process. As a comparison, the electrospun fibers showed a higher crystallinity than the melted samples and a reduced crystallinity than the sample produced by the solution casting method.^[53]

In 2017, Datta et al. reported vertically aligned nylon 11 nanowires produced using an anodized aluminum oxide (AAO) module.^[54] They found that the nylon 11 nanowires were self-poled and that the molecular conformation was in pseudohexagonal γ -phase conformations. When the nylon 11 nanowires were under a small amplitude of displacement (0.5 mm) and low frequency vibrations (5 Hz), they could generate a V_{oc} of 1 V and a 100 nA of I_{sc} . Additionally, the fabricated nylon 11 nanowires also exhibited stability at a high temperature of 150 °C and after $\approx 270\,000$ working cycles. The experimental data are shown in **Figure 6**. The investigation showed that odd nylon-based nanowires can be used as nanogenerators along the thickness direction due to their self-poled piezoelectricity, relatively large output, and high temperature stability. In 2019, Choi et al. produced self-poled nylon 11 nanowires using an assisted gas flow on top of the AAO template during the template-wetting process. Eventually, the nanowire arrays were used to fabricate a triboelectric generator (TEG) with the principle of motion-related surface charge transfer between the nylon nanowire and the contact metal. The nylon nanogenerator exhibited enhanced output performance with a peak V_{oc} of ≈ 110 V and a J_{sc} (short circuit current density) of $\approx 3.8\,\mu\text{A cm}^{-2}$.^[55]

In 2019, Yanaka et al. investigated the relationship between the crystal structure and ferroelectricity of the spin-coated δ -phase nylon 11 films and γ -phase nylon 12. They found that ferroelectric responses were mainly attributed to the lower degree of hydrogen bond in the twisted chain structure in nylon nanofibers.^[56]

In 2020, Anwar et al. fabricated piezoelectric δ -phase nylon 11 nanofibers by ES nylon 11 in trifluoroacetic acid solution.^[57] They found that the δ -phase nylon nanofibers exhibited the highest electrical activity, generating up to 6 V under a mechanical vibration. The finite element simulation results confirmed that the introduction of porosity structure inside the fiber is a feasible way to increase the voltage output.

2.6. Electrospun PBLG Nanogenerator

PBLG possesses a rod-like helical structure with a large number of hydrogen bonds along the main carbon chain direction.^[29] The backbone of the PBLG material aligns in an antiparallel direction along the main chain of the PBLG polymer, demonstrating its piezoelectric properties only along the shear direction. The dipole moments of individual hydrogen bonds can cause enormous dipole moments in the material, which is attributed to the piezoelectric properties of the PBLG material along the shear direction. However, extensive research effort has been devoted to aligning the hydrogen dipoles and producing the piezoelectricity along the thickness direction of PBLG films.^[58–60] In 2008, Farrer et al. polymerized a PBLG solution with MMA (methyl methacrylate) monomers to produce a PBLG-MMA composite with thickness-oriented polarization. During the fabrication process, a PBLG-MMA solution was placed into an AAO module to produce a PBLG-MMA composite. Next, the composite sample was poled along the thickness direction of the PBLG-MMA film using a corona charger to obtain a piezoelectric coefficient of $23\,\text{pC N}^{-1}$. However, this process is relatively complicated, and it is difficult to control the content of the PBLG material.

In 2011, Farrer et al. fabricated PBLG nanofibers using the FFES method.^[41] They found that piezoelectricity can be produced along the dipole direction (d_{33}) with only one step of ES. They also indicated that during the ES process, the shear force and electric field interacted synergistically on the PBLG nanofiber. This interaction enabled dipoles (hydrogen bonds) inside the PBLG material to align along the electric field direction

and exhibit piezoelectric properties along the dipole direction (d_{33} direction). This was for the first time that the piezoelectricity was produced along the d_{33} direction in PBLG nanofibers. From a bending experiment, they also confirmed that the electrical signal generated from the PBLG nanofiber bundle was mainly from the piezoelectric properties rather than the electrostatic charges in the material. Most importantly, they found that PBLG exhibited a large piezoelectric coefficient of 20 pC N^{-1} in the d_{33} direction even after a thermal treatment at 100°C for more than 24 h.

2.7. Electrospun PLLA Nanogenerators

Compared with normal PVDF piezoelectric polymers, a PLLA polymer film possesses piezoelectric properties along the shear direction after a mechanical stretching process without poling. The piezoelectric properties of PLLA nanofibers are mainly derived from C=O dipoles, which have a 125° angle related to the main carbon chain of PLLA.^[61] In 2011, Tajitsu studied PLLA fibers for a biomedical tweezers application.^[62] During the experimental process, two rectangular Al electrodes were deposited on the upper surface of a PLLA fiber (100 μm in diameter) through evaporation method. Due to the piezoelectric properties along the shear direction, a strain was generated at the tip of the PLLA single fiber-based cantilever structure under a sinusoidal electric field. Based on this effect, they designed a fiber tweezer using a pair of PLLA fiber-based cantilever structures and demonstrated that the PLLA fiber tweezer could successfully remove a thrombosis sample from a vessel. This investigation demonstrated a possible biological cell application for spun PLLA fibers. Due to its helical structure, the dipole components of the C=O group will cancel each other out. The parallel dipole component along the main carbon chain is aligned antiparallel with the other components. Thus, the whole structure only shows piezoelectric properties along the shear direction.

In 2016, Y. Tajitsu fabricated wearable PLLA fibers using an extrusion-based high-temperature drawing process to control a humanoid robot.^[63] He demonstrated that the plain, twill, and satin weaves of the spun PLLA fibers produced electrical signals during the bending process of PLLA fibers. Finally, he demonstrated that these woven PLLA fibers were able to control the arm movement of a humanoid robot.

In 2017, Zhao et al. investigated electrospun PLLA fibers for diagnostic sensor applications.^[64] As the data shown in **Figure 7**, a single electrospun PLLA nanofiber generated an output voltage of $\approx 15 \text{ mV}$ (peak-to-peak voltage). They believed that the ES process may align the C=O dipoles in the helical chain of PLLA nanofibers from the antiparallel direction to the parallel direction; the above allowed the nanofiber to show piezoelectric properties along the dipole direction (d_{33}). The measured piezoelectric coefficient d_{14} of the electrospun PLLA nanofibers was $\approx 2.25 \text{ pC N}^{-1}$. This demonstrated for the first time that piezoelectricity was generated along the dipole direction of a PLLA material. More importantly, it was found that a single PLLA fiber-based diagnostic sensor can be used to detect a blood pulse signal. This investigation demonstrated that the PLLA nanofiber could show piezoelectric properties along the dipole direction using the ES method.

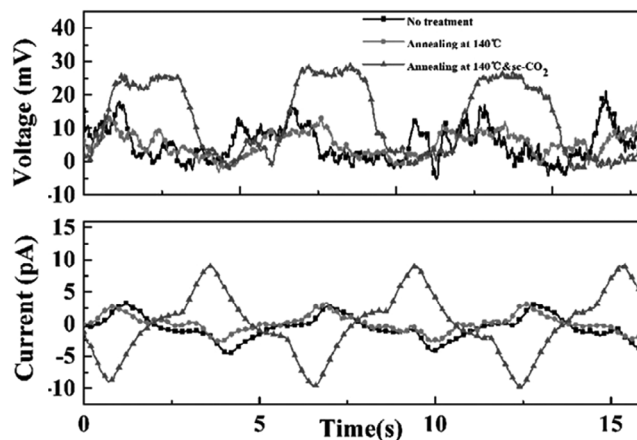


Figure 7. Output voltage and current of PLLA nanofibers in the push–release testing process, where the PLLA nanofibers were treated under various processing conditions (as-spun (solid square), annealed (solid circle), annealed and supercritical CO_2 treatment (solid triangle)).^[64] Reproduced with permission.^[64] Copyright 2017, Wiley-VCH.

In 2017, Sultana et al. fabricated PLLA nanofiber membranes using the FFES method producing the material with a piezoelectric coefficient d_{33} of $\approx 3 \text{ pm V}^{-1}$. The flexible piezoelectric bio-electronic skin was made by sandwiching the PLLA fibers between two layers of polydimethylsiloxane using polyester fabrics coated with Ni-Cu-Ni as electrodes. The device was used to measure the fine movements of the muscles of the internal organs, including the esophagus, trachea, and joint movements.^[65]

3. Characterization of the Electrospun Nanofibers

3.1. Scanning Electron Microscopy

In 2007, Zheng et al. fabricated PVDF nanofibers using the FFES method.^[67] They found that the average fiber diameters were affected by both temperature and solution concentration during the ES process. If the PVDF nanofibers were fabricated using DMF/acetone (8:2) solution in different concentrations from 15% to 20% at 50°C , the diameters of the electrospun PVDF nanofibers changed from 150 to 340 nm and then to 680 nm. Similarly, if a concentration of 15% DMF/acetone (7:3) solution was used to produce PVDF nanofibers at room temperature, a fiber diameter of 1 μm could be obtained after the fabrication process.

In 2018, Wang et al. fabricated PVDF-TrFE nanofibers using the FFES method.^[45] As the SEM images shown in **Figure 8**, they found that the diameters of the PVDF-TrFE fibers varied from ≈ 300 – 600 nm if the PVDF-TrFE in DMF/acetone (6:4) solutions with various concentrations from 15% to 20% were used during fabrication. Similarly, for PLLA nanofibers, Zhao et al. also found that the fiber diameters varied from $\approx 220 \text{ nm}$ to $\approx 1 \mu\text{m}$ if the concentration of a PLLA/DCM solution was changed from 8% to 10%.

In 2011, Farrar et al. fabricated PBLG nanofibers using a 10% PBLG in DCM solution.^[41] They found that the diameters of electrospun PBLG fibers that were produced by the FFES method were ≈ 0.5 – $1 \mu\text{m}$.

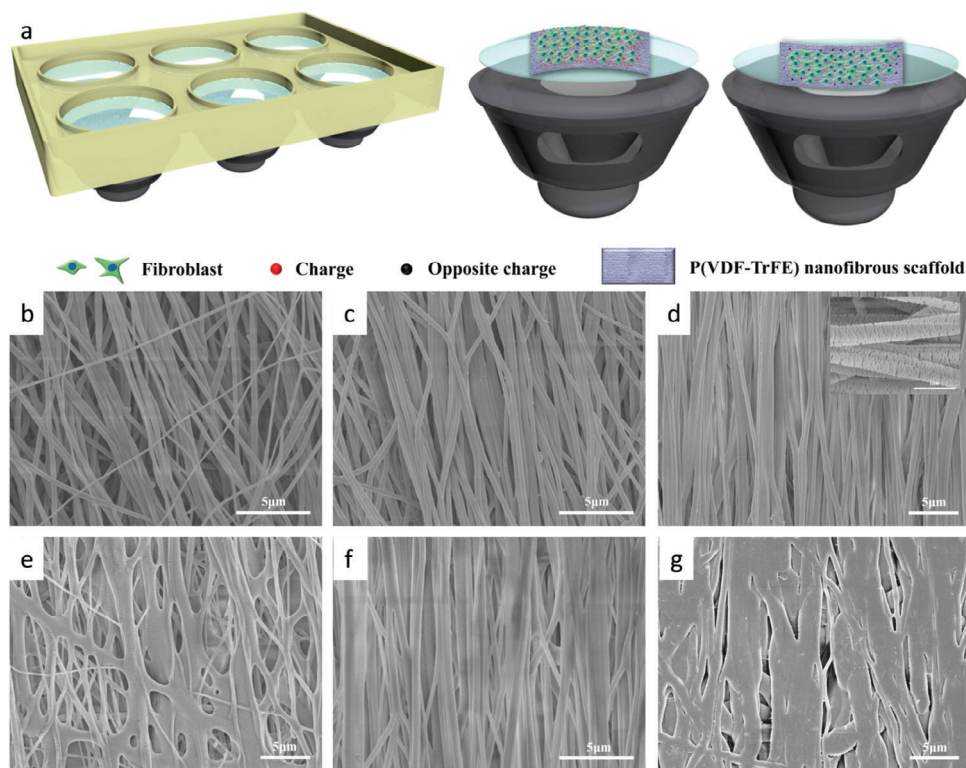


Figure 8. a) Schematic drawing of the experimental setup where the speaker is utilized to excite the piezoelectric properties of electrospun PVDF-TrFE scaffolds on flexible bottom culture plate. b) Scanning electron microscopy (SEM) images of electrospun PVDF-TrFE nanofibers fabricated under varying solution concentrations and various collecting distances after annealing: b) 15%, 10 cm c) 18%, 10 cm d) 20%, 10 cm e) 20%, 5 cm f) 20%, 15 cm g) 20%, 10 cm, pressed. The insets are the higher magnification images of the PVDF nanofiber surface morphologies.^[46] Reproduced with permission.^[46] Copyright 2017, Elsevier Ltd.

3.2. XRD

XRD patterns are normally employed to analyze the structural conformations of polymer materials. In 2007, J. Zheng also found a distinct β phase at 2θ of 20° , which corresponds to the β -phase conformation of a PVDF-TrFE material.^[67] Wang et al. found similar information for electrospun PVDF nanofibers fabricated using the FFES method.^[45] This means that the ES process can greatly benefit the formation of the polar β phase due to the shear force in the ES process, which can stretch the fiber, reducing the α phase and obtaining the β phase. Previously, Ma et al. indicated that by using the FFES method, the strong electric fields during the ES process stretched the PVDF nanofibers and changed PVDF from a nonpolar α phase to a polar β phase.

Similarly, for electrospun PLLA nanofibers, Zhang et al. found that the XRD diffraction peak at 2θ of 16.5° corresponded to the α phase crystal conformation of a PLLA material. The authors also implied that the shear force during the ES process insufficiently stretched the PLLA nanofibers, causing a relatively low piezoelectric coefficient d_{14} in the electrospun PLLA nanofibers.^[68]

Farrar et al. studied the α -helical structure of electrospun PBLG nanofibers and found its dipole orientation direction is associated with the main carbon chain of PBLG material. The XRD diffraction patterns in **Figure 9** indicate “equatorial reflections assigned to a 13 Å spacing and meridional reflections assigned to 5.4 Å spacing, which corresponds to the inter-helix distance and

the helical pitch along the helical axis, respectively.”^[41] The distinct and perpendicular diffraction patterns for the XRD diffraction data implied that the electrospun PBLG nanofibers were aligned in the α -helical conformation and confirmed that their helical axis was aligned along the same direction as the PBLG nanofiber axis.

3.3. Piezoelectric Coefficient Measurement

The piezoelectric coefficient is a critical parameter to evaluate the piezoelectric properties of piezoelectric materials. For most of the previous studies on electrospun piezoelectric polymer nanofibers, the piezoelectric coefficients of the material were seldom measured. Previously, we utilized a lab-designed setup to measure the piezoelectric coefficients of d_{31} and d_{14} for the piezoelectric polymers.^[12,32,45] The lab-designed setup consisted of a shaker and a load cell used to generate and measure the mechanical force and a lock-in amplifier used to measure the generated electrical signal from the measurement sample. For electrospun PVDF-TrFE nanofibers, Wang et al. found that the aligned electrospun PVDF-TrFE nanofibers fabricated with the FFES method generated very little piezoelectricity.^[45] Therefore, the nanofibers need to be poled further to show piezoelectric properties along the fiber spinning direction. The measured piezoelectric coefficient

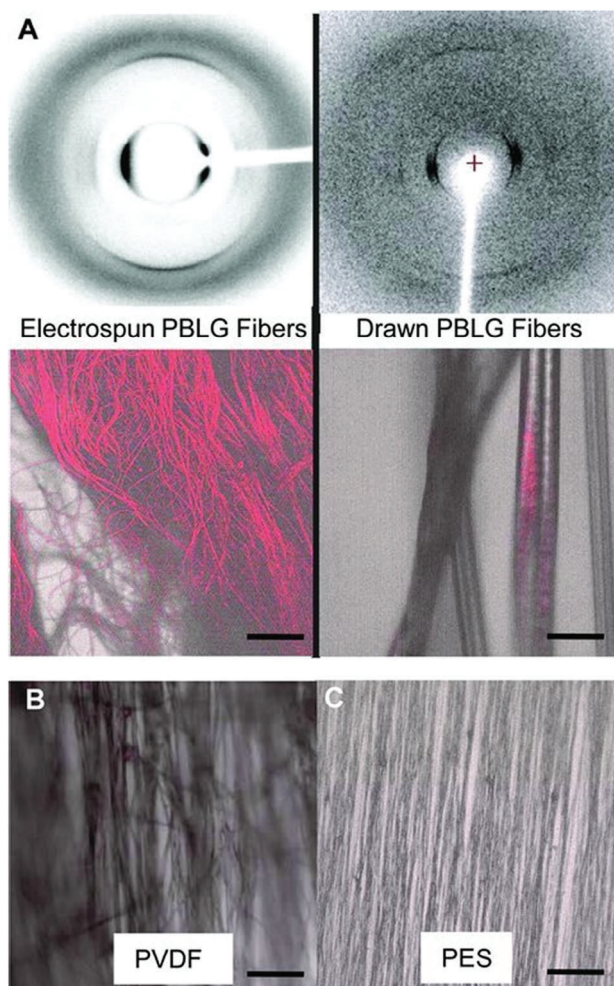


Figure 9. A) Comparative X-ray diffraction patterns and second harmonic generation microscopy images of electrospun (left panels) and drawn (right panels) PBLG fibers. Second harmonic generation microscopy images of B) electrospun PVDF and C) poly(ethersulfone) fibers. Scale bar = 100 μm .^[41] Reproduced with permission.^[41] Copyright 2011, Wiley-VCH.

cient d_{31} for the electrospun PVDF-TrFE nanofibers was 15 pC N^{-1} , which was close to that of a stretched PVDF-TrFE film.

In 2012, Sencadas et al. successfully produced a single PLLA nanofiber on an Au-coated glass substrate using the FFES method and measured its piezoelectric coefficient d_{14} (15 pC N^{-1}).^[69] As shown in Figure 10, D. Farrar measured the piezoelectric coefficient d_{33} of electrospun PBLG fibers and it reached a value of $\approx 20 \text{ pC N}^{-1}$, which was much larger than most biopolymer films.^[41] They also found that the piezoelectric properties of the electrospun PBLG nanofibers did not deteriorate even after incubating at 100 $^{\circ}\text{C}$ for 24 h, which is uncommon for piezoelectric polymer materials. Electrospun PBLG nanofibers show great promise for future sensors and nanogenerator applications.

In 2019, Zhang et al. also measured the piezoelectric coefficient d_{14} of electrospun PLLA nanofibers using the lab-designed experimental setup and found that the d_{14} of the PLLA nanofibers were $\approx 2.25 \text{ pC N}^{-1}$, which was much lower than that of a stretched PLLA film (10 pC N^{-1}).^[68] Due to the advantages

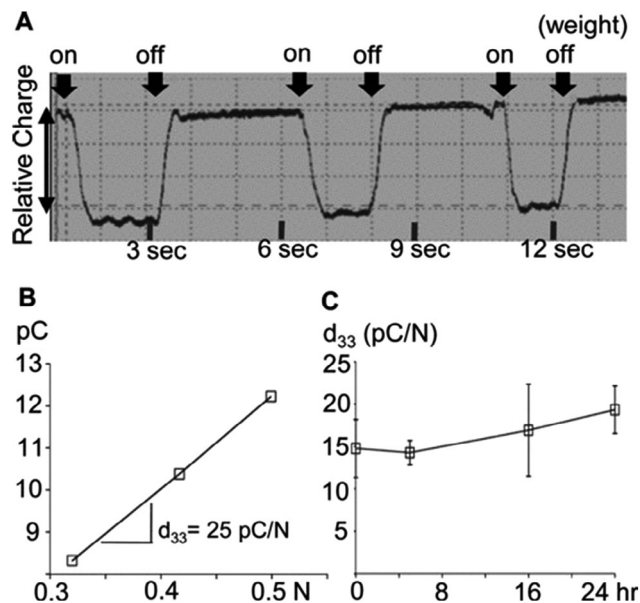


Figure 10. A) Charge accumulation profile during the quasistatic d_{33} test, B) charge versus force plot for the dynamic d_{33} test, and C) the d_{33} thermal stability of the electrospun PBLG fibers.^[41] Reproduced with permission.^[41] Copyright 2011, Wiley-VCH.

of relatively large piezoelectric coefficient, ease of fabrication, and biodegradability, electrospun PLLA nanofibers show great promise in tissue engineering, biosensor, and wearable device applications. As explained in the previous paragraph, Zhang et al. measured the piezoelectric coefficient d_{31} of the electrospun PVDF nanofibers and they found that the PVDF nanofibers using the NFES showed the largest piezoelectric coefficient d_{31} of 0.4 pC N^{-1} , which is significantly higher than that of the PVDF NFs fabricated using the FFES method.^[42]

3.4. Parameters That Affect the Properties of Nanofibers

3.4.1. Solvents

The selection of solvent determines the conformation of dissolved polymer chains, ease of generating fibers, and the evaporation rate of solvent.^[70,71] Basically, the generation of nanofibers from ES system arises from the entanglement of the partially oriented, expanded conformation of polymer chains. The solvent with higher dissolvability and the capability to yield entanglement of polymer chains would be suitable for this application. From this point of view, the selection of solvents is highly dependent on the type of polymers, molecular weight, and ES system, which also need to be studied experimentally.

Previously, Chang et al. utilized a mixture solvent of dimethylformamide (DMF), acetone, and fluorosurfactant (60:40:3) to produce PVDF nanofibers using the NFES method.^[38] During this process, Acetone can accelerate the evaporation of the DMF in this short period. From D. Farrar's paper, the solvent chosen for the fabrication of the PBLG nanofibers is DCM, which has relatively high volatility compared with other solvents, such as DMF, MEK, and HFIP. However, due to the high volatility of DCM, the

PBLG solution can easily get coagulated at the needle tip. Sencadas et al. utilized a mixture solvent of DMF and DCM (3/7) to obtain a 10 wt% PLLA solution and then electrospun it to obtain PLLA nanofibers. Other researchers have also used various ratios of mixed chloroform and DCM solvent, including 95:5, 90:10, 80:20, and 70:30 to fabricate a 4% PLLA solution and compared the electrospun PLLA fibers in various ES conditions.^[27]

3.4.2. Critical Entanglement Concentration C^*

Normally, electrospun polymer nanofibers can be produced using moderately concentrated polymer solutions. In a dilute solution, there is no significant polymer chain interaction and chains exist in separate conditions, thereby no fibers can be formed during the ES process. Based on Huggins equation, the relationship between concentration and viscosity in a polymer solution can be expressed as^[72]

$$\eta_{sp} = [\eta]c + k'[\eta]c^2 + \dots \quad (1)$$

where η_{sp} is the specific viscosity, k' is a constant, and c is the solution concentration. In a dilute polymer solution, the Simha-Frisch parameter $[\eta]c$ is usually less than 1. For a semidilute polymer solution, the solution concentration c is between an overlap concentration c^* and a critical entanglement concentration c_e . Thus, polymer chains begin to interact with each other as the solution concentration increases. Therefore, Dobrynin and his coworkers^[73] predicted that the viscosity of the semidilute solution increases with the square root of the concentration as follows.

$$\eta_{sp} \approx c^{1/2} \quad (2)$$

Theoretical predictions by Dobrynin et al. for an entangled semidilute solution further suggest that the specific viscosity $\eta_{sp} \approx c^{3/2}$. Therefore, sufficient chain entanglements are required to generate polymer nanofibers during the ES process. Essalhi and Khayet found a critical tangle concentration in the PVDF polymer solution, which is $\approx 18.5\%$.^[74] However, for a polymer solution, this critical concentration is directly proportional to the molecular weight (MW) of the polymers. This means a high molecular weight polymer needs a lower concentration to obtain the chain overlap. According to our earlier research, for a PVDF polymer solution (MW: 530 000), the critical concentration of the entangled solution is 20%, which is very close to that of the Essalhi result. It can be concluded that the critical concentration c^* can be changed for different ES methods and ES conditions.

3.4.3. Electric Field

The force generated by the electrostatic charges on the solution at the tip of the needles leads to the formation of the Taylor cone. The instability state of the Taylor cone can be broken up as the electric pressure of polymer solution exceeds the surface tension force. Subsequently, polymer nanofibers are generated through this process. Hohmann et al. studied the stability of electrically forced jet in the ES process and found that at a critical electrical field strength, the instability was completely suppressed, and

the critical electric field E_∞ can be expressed using the following equation.^[75]

$$E_\infty = \sqrt{\frac{2\pi\gamma}{r_0(\epsilon - \bar{\epsilon})}} \quad (3)$$

where γ is the surface tension of the polymer solution, r_0 is the needle diameter, ϵ and $\bar{\epsilon}$ are the dielectric constants of the polymer solution and air, respectively. Ueberreiter et al. theoretically predicted that the surface tension of polymer solutions generally decreases with molecular weight without considering concentration.^[76]

3.4.4. Gap Distance

The distance L (from the needle tip to the collector) determines the strength of electric field and the evaporation rate of solvents before they reach a collector. In an experimental process, the NEFS method normally has a distance of 200 to 500 μm , which is very short compared to that of the FFES method. This small gap may cause wet fibers, that fuse them together on the collector. Meanwhile, when the gap distance L is reduced, the electric field applied to the sample will increase dramatically. Depending on the feed rate and other selected process parameters, a higher electric field will produce thinner fibers but possibly develop an unstable Taylor's cone.^[26] To solve these problems, solvents with high viscosity should be chosen for NEFS to obtain continuous PVDF nanofibers.

PVDF has various phase conformations including α phase, β phase, and γ phase depending on the process conditions.^[77,78] However, it is known that the FFES process is similar to the polymer stretching process which is necessary for generating piezoelectricity in PVDF-based polymer film. From our previous research results, the electrospun PVDF nanofibers by the FFES process exhibit very little piezoelectricity. We believed that as shear force and electric field coexist in fiber spinning condition, shear force will dominate the polymer alignment. Thus an extra poling process is necessary to align all the dipole along the thickness direction to show piezoelectricity on the PVDF fibers eventually.^[9] In 2010, Chang et al. successfully fabricated PVDF nanogenerator using the NFES method and proved that the energy conversion efficiency of PVDF nanofibers (12.5%) is an order of magnitude higher than normal PVDF films ($\approx 1\%$),^[13] which mainly resulted from the increasing piezoelectricity in the PVDF nanofibers by the NFES method. For the FFES method, if the tip distance L continues to increase while the other parameters stay the same, the diameter of the fiber will be reduced. In fact, the process of generating fibers from the tip of the needle to the collector makes it possible to stretch the polymer fibers and this will improve the mechanical strength of polymer fibers. The ES process is therefore an integrated stretching and poling process, especially for piezoelectric materials with a dipole parallel to the helical chain, including PLLA and PBLG.^[10]

According to previous publication results, the gap distance L of 15 cm is an optimized condition for obtaining uniform PLLA nanofibers with a 10% solution concentration. Moreover, the gap distance L can also determine the stretching ratio of nanofibers in the FFES method. Sencadas et al. successfully produced a

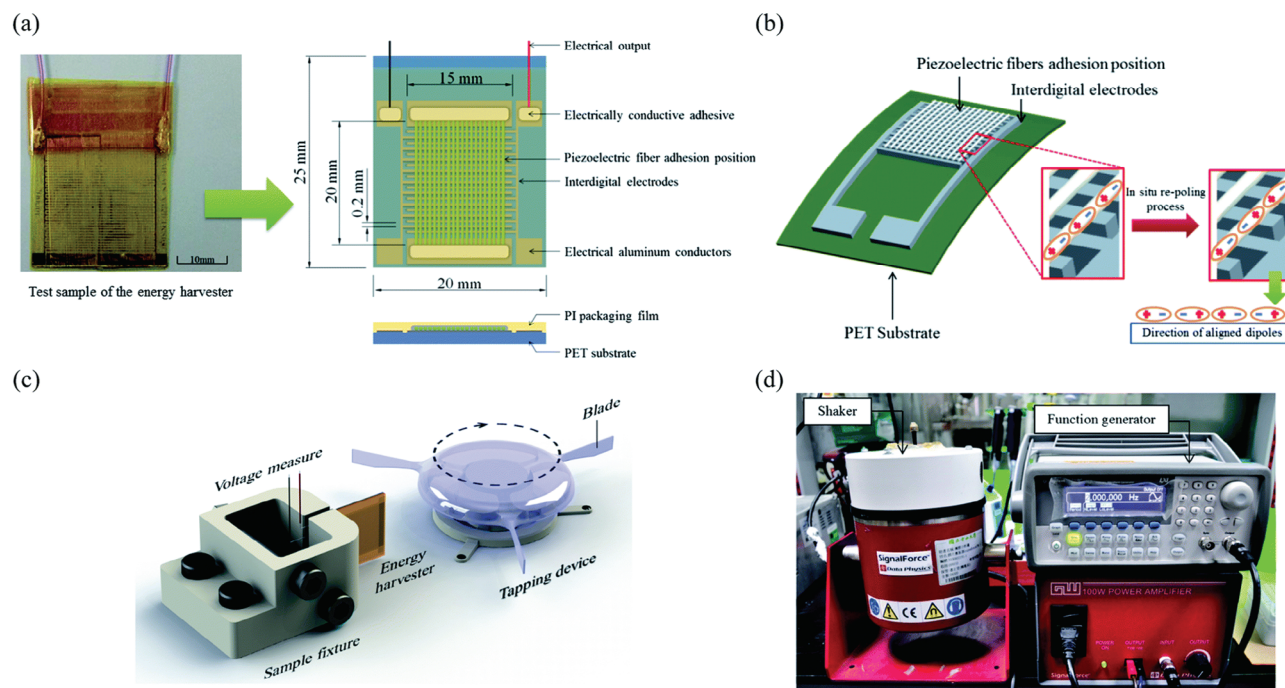


Figure 11. a) Schematic representation and test sample of the energy harvester. b) A schematic diagram of fibers with in situ re-poling process. c) Equipment used to measure the performance of energy harvester. d) Shaker system for energy harvesting with an insect wing.^[79] Reproduced with permission.^[79] Copyright 2014, Royal Society of Chemistry.

single PLLA nanofiber on a Au-coated glass substrate using the FFES method and measured its piezoelectric coefficient d_{33} (15 pC N^{-1}).^[17]

The gap distance L in the ES setup can determine the strength of the electric field as well as the evaporation rate of solvents before it reaches the target. The NEFS normally has a gap L from 200 to 500 μm . Meanwhile, when the gap distance L is reduced, the electric field applied to the sample can increase dramatically. A high electric field during this process can produce a large poling field and benefit the alignment of the dipoles in piezoelectric polymers, especially PVDF-related polymers with the dipoles perpendicular to the main chain of the polymer.^[38]

For the FFES method, the fiber-generating process from the needle tip to the rotating collector benefits to stretch the polymer fibers and increase the mechanical strength of the polymer fibers. For the aligned PVDF nanofibers fabricated using the FFES method (rotating collector), the direction of the electric field and the main carbon chain of the polymer is parallel with each other. Therefore, it can induce very weak piezoelectric properties in the PVDF nanofibers. For randomly aligned PVDF nanofibers, the electric field is always perpendicular to the polymer nanofibers during the ES process. However, the residual solvent in the fibers can deteriorate the alignment of dipoles along the thickness direction of the PVDF fibers, which can significantly reduce the piezoelectricity of the nanofibers. In addition, for piezoelectric materials with dipoles parallel to the main carbon chain, including PLLA and PBLG, the FFES process can benefit the orientation of the dipoles along the electric field direction and induce piezoelectric properties along the d_{33} direction of the materials.^[41]

4. Energy Harvesting and Sensor Applications Using Electrospun Polymer Nanofibers

In 2014, Pan et al. fabricated PBLG nanofibers using the NFES method with a cylindrical collector for electric energy harvesting applications.^[79] Their process yielded PBLG nanofibers with highly oriented α -helical structures. The fabricated PBLG nanofibers were adhered to interdigital electrodes (IDEs) for testing. The output voltage of the electrospun PBLG nanofibers was 33 mV, and the maximum output power of the device reached 138 pW (Figure 11). Finally, the electrospun PBLG nanofibers were adhered on cicada wings to harvest electric energy from wing vibrations. They obtained a V_{oc} of 14.2 mV and an I_{sc} of 19 nA at a cicada wing vibration of 30 Hz.

In 2013, J. Nunes-Pereira investigated electrospun PVDF, electrospun PVDF-TrFE, and PVDF/BTO nanofibers for energy harvesting applications.^[80] These nanofibers were deposited on IDEs using the FFES method, and the output performance was measured using a cantilever beam. They found that the electrospun PVDF, PVDF-TrFE, and PVDF-TrFE/BaTiO₃ nanofibers exhibited output voltages of 140, 110, and 100 mV, respectively.

In 2017, Lang et al. investigated electrospun PVDF nanofibers for acoustic sensor applications.^[81] They fabricated PVDF nanofibers with a variety of fiber dimensions using varied PVDF solution concentrations. They found that the PVDF fibers with the smallest fiber dimension exhibited higher piezoelectric potential than the other fibers, and the uniform PVDF fibers without beads could show the largest output voltage. Finally, they demonstrated that electrospun PVDF nanofibers could be used

to detect a sound to produce a similar signal to that from a commercial acoustic sensor.

In 2019, He et al. utilized electrospun PVDF nanofibers as a piezoelectric nanogenerator and electrodeposited them on a patterned polyaniline electrode-based electrochromic supercapacitor (ESC) as an energy harvesting and energy storage combined device. The color change of the ESC device can be used to monitor the palm impaction process.^[82]

In 2022, Nikbakhtnasrabadi et al. printed an interdigitated circuit (IDC) on an electrospun PLLA nanofiber membrane to make a pressure sensor. The design of the device isolated the charge generated from the IDC and PLLA nanofibers using a polyimide (PI) film sheet. The device exhibited excellent durability with a high sensitivity of 0.035 k Pa^{-1} and 1200 Hz kPa^{-1} . The sensor was also integrated in a bandage to promote the regeneration of cells.^[66]

5. Summary and Future Prospects

In summary, it is concluded that piezoelectricity can be generated using ES methods. Depending on the symmetric structure of piezoelectric polymers, ES method may have various impacts on the piezoelectric properties of electrospun polymer nanofibers. For PVDF-related polymers with the dipoles perpendicular to the polymer surface, the NFES method can have a large effect on the piezoelectric property of polymer nanofibers. For the FFES methods with a rotating collector, because the direction of the electric field is parallel to the main carbon chain of the polymer, this process may have less effect on the piezoelectric properties of PVDF-related polymer nanofibers. However, for biopolymers with the dipoles parallel to the main carbon chain, such as PLLA and PBLG, the FFES method can benefit the orientation process of the polymer nanofibers, which can help produce the piezoelectricity in the d_{33} direction and shear direction (d_{14}) of polymer nanofibers. Due to the biocompatibility and relatively large piezoelectric coefficients, these electrospun biopolymer nanofibers show great promise in wearable electronic devices and wound healing applications. For the relatively large piezoelectric coefficient produced in PVDF nanofibers by the FFES method, these nanofibers show enormous potential in nanogenerator, sensor, and cell proliferation applications.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

far-field electrospinning, near-field electrospinning, poly(L-lactic acid) nanofibers, poly(vinylidene difluorine) nanofibers, sensor applications

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